

Domestic and farm-animal exposures and risk of non-Hodgkin lymphoma in a population-based study in the San Francisco Bay Area



Objective To assess the association between animal exposures and non-Hodgkin lymphoma (NHL). **Methods** Exposure data were collected from 1,591 cases and 2,515 controls during in-person interviews in a population-based case-control study of NHL in the San Francisco Bay Area. Odds ratios (ORs) and 95% confidence intervals (CIs) were adjusted for potential confounders. **Results** Pet owners had a reduced risk of NHL (OR=0.71, CI=0.52–0.97) and diffuse large-cell and immunoblastic large-cell (DLCL; OR=0.58, CI=0.39–0.87) compared with those who never had owned a pet. Ever having owned dogs and/or cats was associated with reduced risk of all NHL (OR=0.71, CI=0.54–0.94) and of DLCL (OR=0.60, CI=0.42–0.86). Longer duration of cat ownership (p-trend=0.008), dog ownership (p-trend=0.04), and dog and/or cat ownership (p-trend=0.004) was inversely associated with risk of NHL. Ownership of pets other than cats and dogs was associated with a reduced risk of NHL (OR=0.64, CI=0.55–0.74) and DLCL (OR=0.58, CI=0.47–0.71). Exposure to cattle for ≥ 5 years was associated with an increased risk of NHL (OR=1.6, CI=1.0–2.5) as was exposure to pigs for all NHL (OR=1.8, CI=1.2–2.6) and for DLCL (OR=2.0, CI=1.2–3.4). **Conclusions** The association between animal exposure and NHL warrants further investigation in pooled analyses.